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International Mathematics Assessments for Schools (2025-2026)

Middle Primary Division Round 2

Time: 120 minutes

Printed Name _____ *Code* _____ *Score* _____

Instructions:

- Do not open the contest booklet until you are told to do so.
- Be sure that your name and code are written on the space provided above.
- Round 2 of IMAS is composed of three parts; the total score is 100 marks.
- Questions 1 to 5 are given as a multiple-choice test. Each question has five possible options marked as A, B, C, D and E. Only one of these options is correct. After making your choice, fill in the appropriate letter in the **answer sheet**. Each correct answer is worth 4 marks. There is no penalty for an incorrect answer.
- Questions 6 to 13 are a short answer test. Only Arabic numerals are accepted; using other written text will not be honored or credited. Some questions have more than one answer, as such all answers are required to be written down in the **answer sheet** to obtain full marks. Each correct answer is worth 5 marks. There is no penalty for incorrect answers.
- Questions 14 and 15 require a detailed solution or process in which 20 marks are to be awarded to a completely written solution. Partial marks may be given to an incomplete presentation. There is no penalty for an incorrect answer.
- Use of electronic computing devices is not allowed.
- Only pencil, blue or black ball-pens may be used to write your solution or answer.
- Diagrams are not drawn to scale. They are intended as aids only.
- After the contest the invigilator will collect the contest paper.

The following area is to be filled in by the judges;
the contestants are not supposed to mark anything here.

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Total Score	Signature
Score																	
Score																	

Middle Primary Division Round 2

Questions 1 to 5, 4 marks each

1. A family has four children. On January 1, 2020, their ages were in the ratio 2:3:4:5 and the sum of their birth years is 8038. What is the sum of the children's ages on January 1, 2026?

（A）42　（B）54　（C）58　（D）62　（E）66

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2. A car and a truck travel in the same direction along a straight highway. At a certain moment, the truck is 1 km ahead of the car. The truck travels at a constant speed of 88 km/h, while the car travels at a constant speed of 100 km/h. What is the distance, in m, between the truck and the car one minute before the car catches up with the truck?

（A）100　（B）120　（C）150　（D）200　（E）240

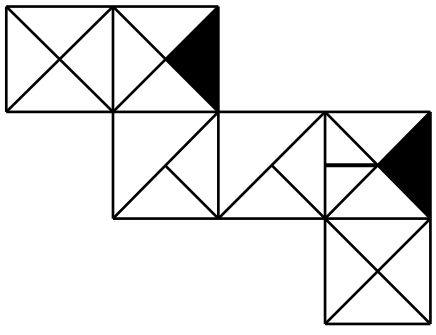
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3. A party had between 41 and 48 attendees. In each dance, every attendee who participates in the dance is paired with exactly one person of the opposite sex. At the end of the party, it is known that each boy had danced with exactly four different girls, and each girl had danced with exactly five different boys. How many girls attended the party in total?

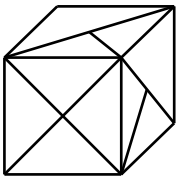
（A）18　（B）20　（C）24　（D）25　（E）Uncertain

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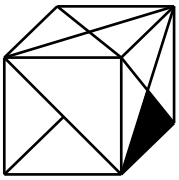
4. Which object among options (A) - (E) is **IMPOSSIBLE** to construct by folding the net that is shown below?



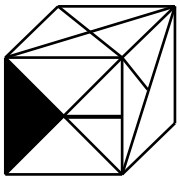
（A）



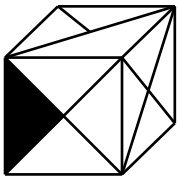
（B）



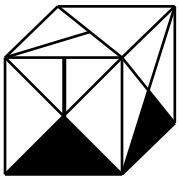
（C）



（D）



（E）



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5. Along the path between Alice's house and Ben's house is a line of flowers consisting of 25 peonies and 25 tulips. One day, Alice walks from her house toward Ben's house, watering the flowers one by one as she encounters them. After watering the 20th tulip, her water runs out, leaving exactly 10 flowers unwatered. The next day, Ben walks from his house toward Alice's house, picking the flowers one by one as he encounters them. After picking the 6th tulip, he decides that he has enough flowers for a bouquet. How many flowers remain along the path?

（A）34　　（B）38　　（C）39　　（D）40　　（E）Uncertain

Questions 6 to 13, 5 marks each

6. The set of positive even integers is divided into groups, as follows:

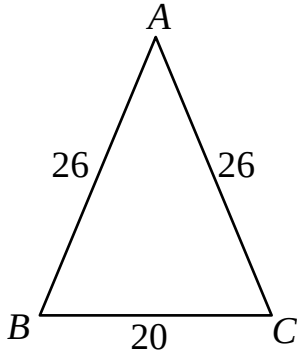
1st group: 2
2nd group: 4, 6
3rd group: 8, 10, 12
4th group: 14, 16, 18, 20
…

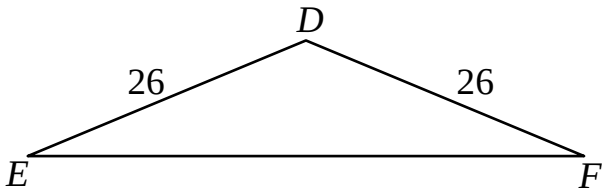
What is the first number in the 100th group?

7. An ant goes out to look for food. After finding food, it immediately returns to its nest and brings back 15 more ants. These ants are still unable to move the food, so half of them return to the nest, and each brings back 10 more ants. They are still unable to move the food, so one-third of the ants return to the nest, and each brings back 8 more ants. Still unable to move the food, one-fourth of the ants return to the nest, and each brings back 6 more ants. Again, unable to move the food, one-fifth of the ants return to the nest, and each brings back 4 more ants. Finally, the ants are able to move the food. How many ants were there in total when the food was moved?

8. What is the positive integer that is equal to 12 times the sum of its digits (when written in base 10)?

9. In triangle ABC, $AB = AC = 26$ cm and $BC = 20$ cm. In triangle DEF, $DE = DF = 26$ cm, where the length of EF is an integer, in cm, with $EF \neq BC$. It is known that the two triangles have the same area. What is the length, in cm, of EF?





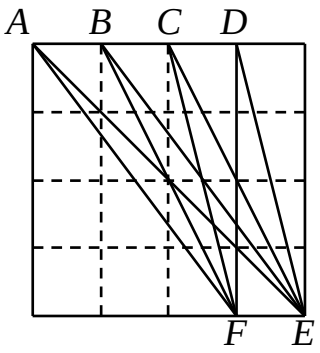
10. Start with the number 2. At each turn, you may either add 4 to the current number or multiply the current number by 3. For example, it is possible to reach exactly 82 in 18 turns as shown below:

$$(2+4)\times 3+4+4+4+4+4+4+4+4+4+4+4+4+4+4+4+4=82.$$

What is the minimum number of turns required to get 82?

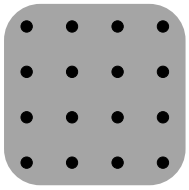
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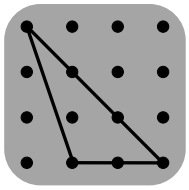
11. As shown in the diagram below, points A, B, C, D, E and F are grid points marked on the 4×4 square.
What is the sum, in degrees, of $\angle EAF+\angle EBF+\angle ECF+\angle EDF$?



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12. As shown in the diagram on the left below, a 4×4 grid is created on a wall by using sixteen nails, with the distance between any two adjacent nails, whether horizontally or vertically, equal to 10 cm. A rubber band is then used to connect three non-collinear points on the grid to form a triangle. The diagram on the right below shows one example of a triangle formed in this manner.

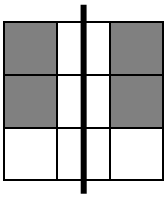




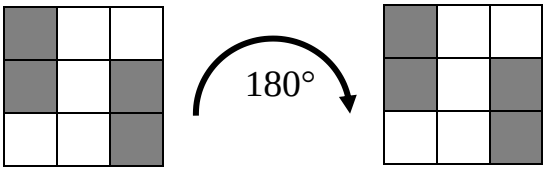
How many different possible values can the area of the resulting triangle formed in this manner take?

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13. As shown in the diagram below, four unit squares of the 3×3 grid are painted gray and the thick line represents one line of symmetry of the 3×3 grid.



As shown in the diagram below, four unit squares of the 3×3 grid are painted gray so that the diagram looks exactly the same when rotated by $180°$. We say that the pattern has $180°$ rotational symmetry.



Now, three unit squares of the following 5×5 grid are painted gray. Ann needs to paint additional unit squares gray so that the resulting shape has at least one line of symmetry and looks exactly the same when rotated by $180°$. If Ann paints the least possible number of squares, what is the sum of the numbers written inside the additional unit squares she paints, excluding the initial three? (Note: the orientation of the numbers written inside the squares is ignored when assessing the symmetry of the shaded pattern.)

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25

ANSWER SHEET of Questions 1 to 13

Questions 1 to 5, 4 marks each

- Questions 1 to 5 are given as a multiple-choice test. Each question has five possible options marked as A, B, C, D and E. Only one of these options is correct.

1.		**2.**		**3.**		**4.**		**5.**	

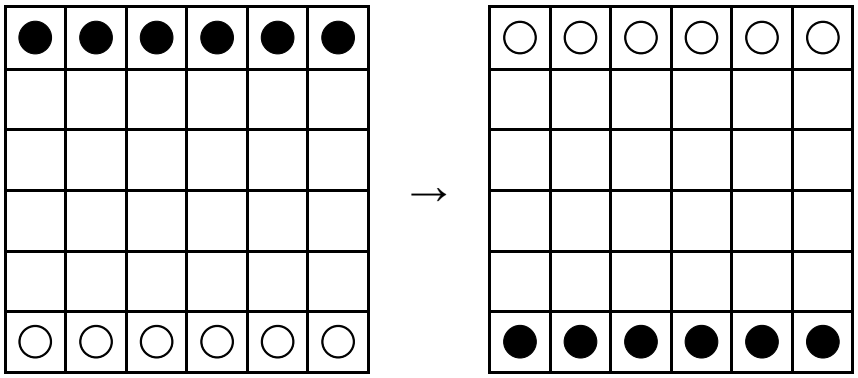
Questions 6 to 13, 5 marks each

- Questions 6 to 13 are a short answer test. Only Arabic numerals are accepted; using other written text will not be honored or credited.

6.		**7.**	ants	**8.**		**9.**	cm
10.	turns	**11.**	°	**12.**		**13.**	

Questions 14 to 15, 20 marks each
(Detailed solutions are needed for these two problems)

14. As shown in the diagram on the left below, there are six identical black circles in the first row of a 6×6 chessboard and six identical white circles in the last row. The circles can move one at a time, horizontally, vertically or diagonally (parallel to the main diagonals) by any number of squares as long as no other circle blocks the path. Black and white circles move alternately. What is the minimum number of moves required to interchange the black and white circles as shown in the diagram on the right below? You must prove that achieving the goal in fewer moves is impossible and provide one sequence of moves that accomplishes the interchange using the minimum number of moves.



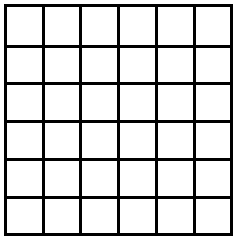
Answer： _____ moves

(Give a sequence of moves that accomplishes the interchange in the next page)

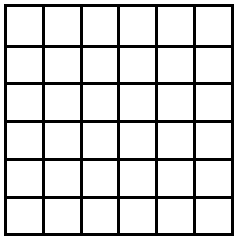
MP 6

(The sequence of moves that accomplishes the interchange of Question 14)

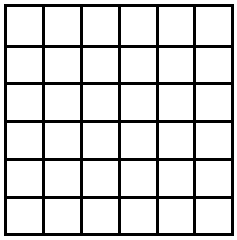
Move 1



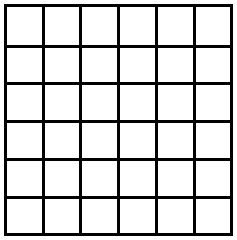
Move 2



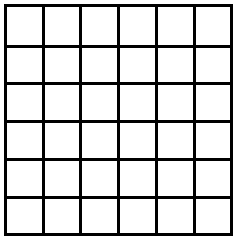
Move 3



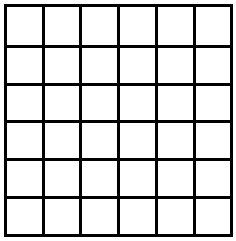
Move 4



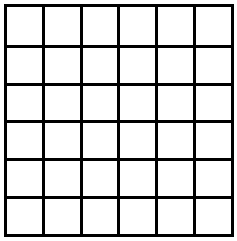
Move 5



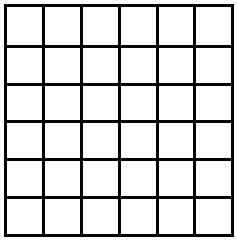
Move 6



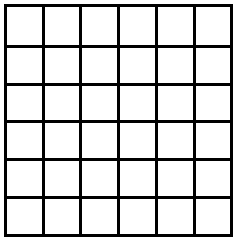
Move 7



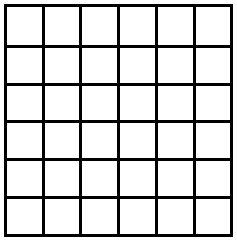
Move 8



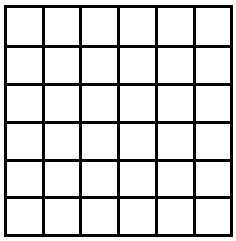
Move 9



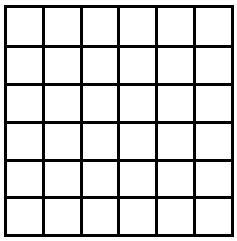
Move 10



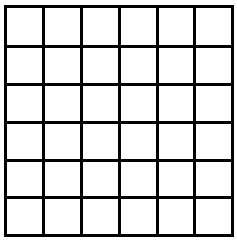
Move 11



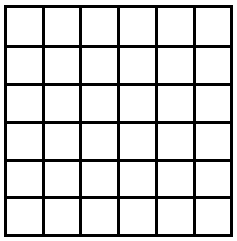
Move 12



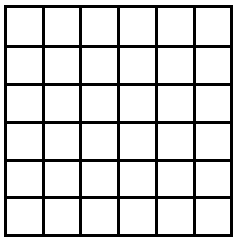
Move 13



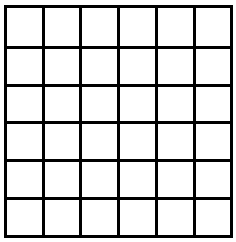
Move 14



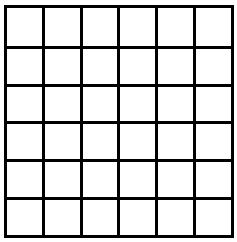
Move 15



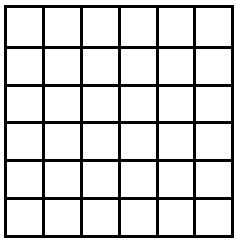
Move 16



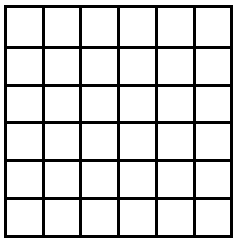
Move 17



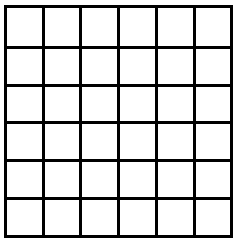
Move 18



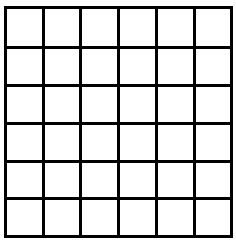
Move 19



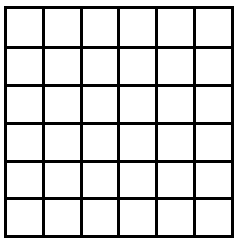
Move 20



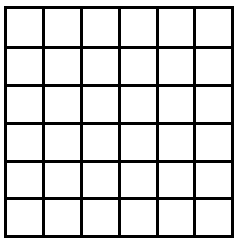
Move 21



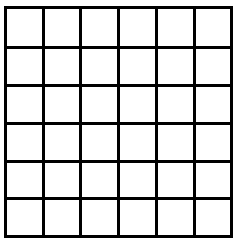
Move 22



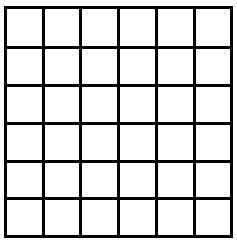
Move 23



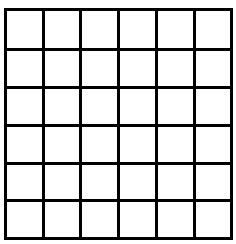
Move 24



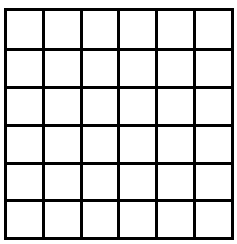
Move 25



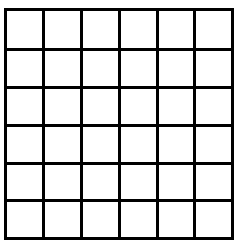
Move 26



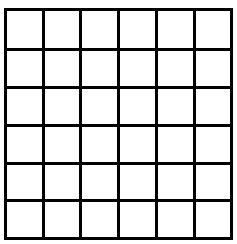
Move 27



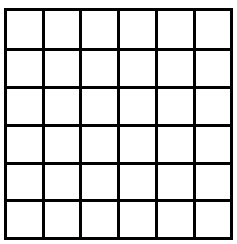
Move 28



Move 29



Move 30



15. In Braille code, a positive integer is represented by the number sign followed by the Braille symbols for each of its digit, as shown in the table below. For example, the number 27 is represented as ⠼⠃⠛, using a total of 10 dots.

Number sign	1	2	3	4	5	6	7	8	9	0
⠼	⠁	⠃	⠉	⠙	⠑	⠋	⠛	⠓	⠊	⠚

Including the number sign ⠼, how many positive integers require exactly 8 dots to represent?

Answer : _____ positive integers

***